Modular Powers and Primality Testing

COMP2711 Discrete Mathematical Tools

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1. Review of Bézout's Identity
2. Fermat's Little Theorem
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4. Optional: Miller–Rabin Primality Testing

1.

Review of Bézout's Identity

Review: Bézout's identity

Definition: Integer linear combination

An **integer linear combination** of integers a, b is a number $ax + by$, where $x, y \in \mathbb{Z}$.

Theorem (Bézout)

For integers a, b not both zero, there exist integers x, y such that

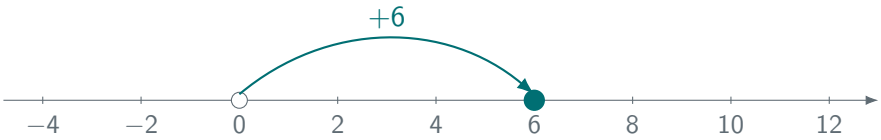
$$ax + by = \gcd(a, b).$$

Interpretation of Bézout

It describes all possible integer linear combinations of a and b.

Example: Which values can $6x + 10y$ attain?

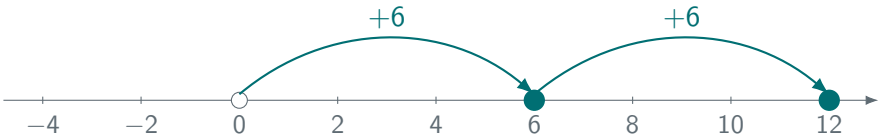
Fix $a = 6$, $b = 10$. Start at 0; add or subtract 6 or 10.



$$6 = 6 \cdot 1 + 10 \cdot 0, \qquad (x, y) = (1, 0).$$

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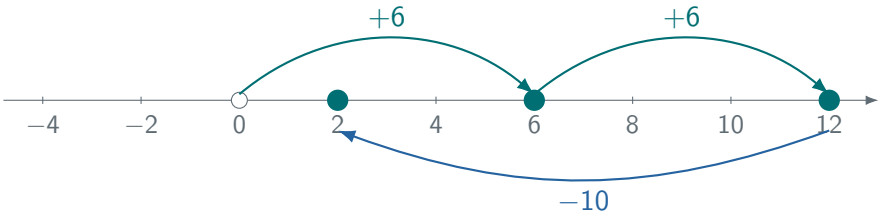
Fix $a = 6$, $b = 10$. Start at 0; add or subtract 6 or 10.



$$12 = 6 \cdot 2 + 10 \cdot 0, \qquad (x, y) = (2, 0).$$

Example: Which values can $6x + 10y$ attain?

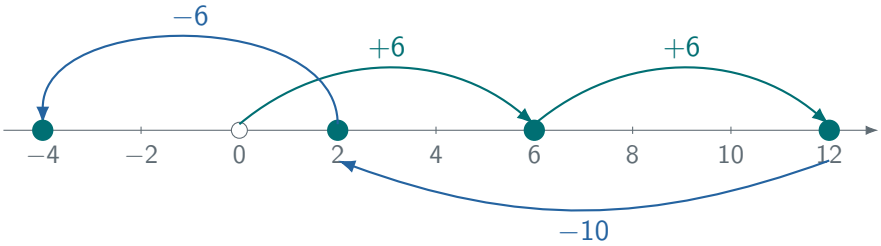
Fix $a = 6$, $b = 10$. Start at 0; add or subtract 6 or 10.



$$2 = 6 \cdot 2 + 10 \cdot (-1), \qquad (x, y) = (2, -1).$$

Example: Which values can $6x + 10y$ attain?

Fix $a = 6$, $b = 10$. Start at 0; add or subtract 6 or 10.



$$-4 = 6 \cdot 1 + 10 \cdot (-1), \qquad (x, y) = (1, -1).$$

Bézout's identity: Integer linear combinations

Fix $a, b \in \mathbb{Z}$, not both zero. Which values can $ax + by$ attain for $x, y \in \mathbb{Z}$?

1. Positive minimum

$\gcd(a, b)$ is the smallest positive value of $ax + by$.

Let $g = \gcd(a, b)$. For any integers x, y,

$$g \mid a \implies g \mid ax, \qquad g \mid b \implies g \mid by.$$

Therefore $g \mid (ax + by)$, so $ax + by$ is always a multiple of g.

No value strictly between 0 and g is attainable.

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2. Attain the GCD

$ax_0 + by_0 = \gcd(a, b)$

Extended Euclid: the backward step. Let $b \neq 0$ and $q = a \operatorname{div} b$. If $sb + t(a \bmod b) = \gcd(a, b)$, substitute $a \bmod b = a - qb$:

$$sb + t(a - qb) = ta + (s - qt)b = \gcd(a, b).$$

The coefficients for (a, b) are therefore $(x_0, y_0) = (t, s - qt)$.

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$ax_0 + by_0 = \gcd(a, b)$

3. All multiples

$k \gcd(a, b), \quad k \in \mathbb{Z}$

To reach any target point $k \gcd(a, b)$:

$$a(kx_0) + b(ky_0) = k \gcd(a, b).$$

2.

Fermat's Little Theorem

Review: Computing modular powers

Compute

$$2^{13} \equiv ? \pmod 5.$$

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Cycles

$$2 \to 4 \to 3 \to 1 \to 2 \pmod 5$$

$$2^{13} = (2^4)^3\, 2 \equiv 2 \pmod 5.$$

Repeated squaring

$$13 = 8 + 4 + 1$$

$$\begin{aligned} 2^{13} &= 2^8\, 2^4\, 2 \\ &\equiv 1 \times 1 \times 2 \pmod 5. \end{aligned}$$

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Can a theorem give a valid exponent reduction without first building the cycle?

Fermat's little theorem

MUST UNDERSTAND Modular powers

If p is **prime** and $1 \leq a \leq p-1$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Fermat's little theorem

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If p is **prime** and $1 \leq a \leq p-1$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Example:

$$3^6 \equiv 1 \pmod{7}.$$

Equivalent all-integer form:

$$a^p \equiv a \pmod{p} \qquad \text{for every integer } a.$$

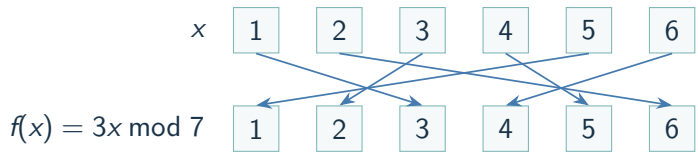
Fermat: Multiplication rearranges remainders

p prime, $1 \le a \le p-1$
$a^{p-1} \equiv 1 \pmod{p}$

Consider $S = \{1, 2, \ldots, p-1\}$ and

$$f : S \to S, \qquad f(x) = ax \bmod p.$$

Example: $p = 7$, $a = 3$.



Fermat: Why outputs are nonzero

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Hint: Since $\gcd(a, p) = 1$, a has an inverse a^{-1} modulo p.

Fermat: Why the map is a bijection

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Suppose $f(x) = f(y)$ for $x, y \in S$. Then

$$\begin{aligned} ax \equiv ay \pmod{p} &\implies a^{-1}ax \equiv a^{-1}ay \pmod{p} \\ &\implies x \equiv y \pmod{p} \implies x = y. \end{aligned}$$

Thus f is injective (i.e., the same output implies the same input).

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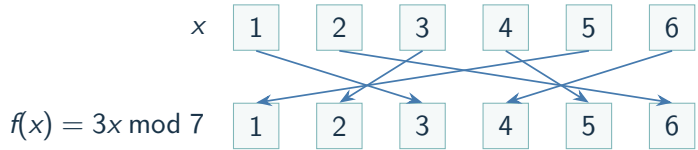
By the pigeonhole principle: An injective function from a finite set to itself is bijective, so f is bijective.

Fermat: Multiply the same remainders

We have proved that f is a bijection.

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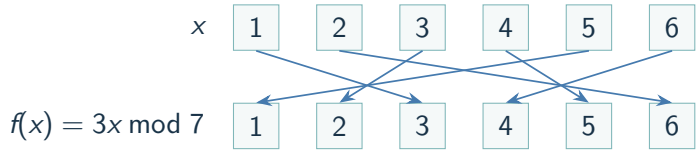


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Now we know:

$$1 \times 2 \times \cdots \times (p-1) = f(1)f(2)\cdots f(p-1).$$

The same product of elements in S, written in a different order.

Fermat: Cancel the invertible product

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$$1 \times 2 \times \cdots \times (p-1) = f(1)f(2)\cdots f(p-1).$$

Substitute $f(x) \equiv ax \pmod{p}$ into the product identity:

$$(p-1)! \equiv (a \times 1)(a \times 2)\cdots(a \times (p-1)) \pmod{p}.$$

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There are $p-1$ factors of a, so

$$(p-1)! \equiv a^{p-1}(p-1)! \pmod{p}.$$

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Every factor $1, \ldots, p-1$ is coprime to p, so $(p-1)!$ is invertible.

Cancel $(p-1)!$:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}.}$$

Corollary: An inverse from Fermat's little theorem

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If p is prime and $1 \leq a \leq p-1$, then

$$a^{p-2} \equiv a^{-1} \pmod{p}.$$

That is, a^{p-2} is an inverse of a modulo p.

Corollary: An inverse from Fermat's little theorem

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That is, a^{p-2} is an inverse of a modulo p.

By Fermat's little theorem,

$$a \times a^{p-2} = a^{p-1} \equiv 1 \pmod{p}.$$

Computing a large power with Fermat

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Write $100 = 16 \times 6 + 4$:

$$3^{100} = (3^6)^{16} 3^4 \equiv 1^{16} \times 81 \equiv \boxed{4} \pmod 7.$$

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Practice: Compute $7^{222} \bmod 11$. $(222 = 22 \times 10 + 2.)$

3.

Euler's Totient Function

Euler's totient

Definition: Euler's totient

For $m > 1$, $\varphi(m)$ counts how many $x \in \{1, \ldots, m-1\}$ satisfy

$$\gcd(x, m) = 1 \qquad \text{(i.e., } x \text{ is coprime to } m\text{)}.$$

Equivalently, it counts how many of these x have an inverse x^{-1} modulo m.

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For any prime p, $\varphi(p) = p - 1$, since $1, 2, \ldots, p-1$ are all coprime to p.

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0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

Example: $\varphi(15) = 8$.

Euler's theorem

Theorem (Euler)

If $m > 1$ *and* $\gcd(a, m) = 1$, *then*

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

The proof is a simple extension of the proof of Fermat's little theorem: use the set of remainders coprime to m.

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If $m > 1$ *and* $\gcd(a, m) = 1$, *then*

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The proof is a simple extension of the proof of Fermat's little theorem: use the set of remainders coprime to m.

For any integer $k \geq 0$,

$$\boxed{a^k \equiv a^{k \bmod \varphi(m)} \pmod{m}.}$$

Using Euler's theorem

Compute $2^{100} \bmod 15$**.**

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We have $\gcd(2, 15) = 1$ and $\varphi(15) = 8$, so $2^8 \equiv 1 \pmod{15}$.

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Since $100 = 12 \times 8 + 4$,

$$2^{100} = (2^8)^{12} 2^4 \equiv 16 \equiv \boxed{1} \pmod{15}.$$

Using Euler's theorem

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Since $100 = 12 \times 8 + 4$,

$$2^{100} = (2^8)^{12} 2^4 \equiv 16 \equiv \boxed{1} \pmod{15}.$$

Can we use the same theorem for $3^{100} \bmod 15$?

Computing Euler's totient: Two facts

1. Prime powers

For any prime p and integer $k \geq 1$,

$$\begin{aligned}\varphi(p^k) &= p^k - p^{k-1} \\ &= (p-1)p^{k-1}.\end{aligned}$$

Explanation: Among $1, \ldots, p^k - 1$, all numbers except the multiples $p, 2p, 3p, \ldots$ are coprime to p^k.

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2. Coprime factors

For positive integers a, b with $\gcd(a, b) = 1$,

$$\varphi(ab) = \varphi(a)\varphi(b).$$

Explanation: By CRT, each pair of remainders (a', b') with $\gcd(a', a) = \gcd(b', b) = 1$ determines exactly one remainder modulo ab, coprime to ab.

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We use $\varphi(1) = 1$.

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Explanation: By CRT, each pair of remainders (a', b') with $\gcd(a', a) = \gcd(b', b) = 1$ determines exactly one remainder modulo ab, coprime to ab.

Computing Euler's totient: General formula

1. Write the prime factorization. For $x > 1$,

$$x = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t},$$

where the p_i are distinct primes and each $e_i \geq 1$.

Computing Euler's totient: General formula

1. Write the prime factorization. For $x > 1$,

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where the p_i are distinct primes and each $e_i \geq 1$.

2. Apply the two facts. The prime-power factors are pairwise coprime.

Euler's totient formula

$$\begin{aligned}\varphi(x) &= \prod_{i=1}^{t} \varphi(p_i^{e_i}) = \prod_{i=1}^{t} (p_i - 1) p_i^{e_i - 1} \\ &= x \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right).\end{aligned}$$

Example: Computing a totient

Compute $\varphi(60)$ from $60 = 2^2 \times 3 \times 5$.

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For two distinct primes p, q,

$$\varphi(pq) = \varphi(p)\varphi(q) = (p-1)(q-1).$$

Example: $\varphi(55) = (5-1)(11-1) = 40$.

Review exercises: Modular powers

1. **Fermat's little theorem.** Compute $5^{100} \bmod 7$.

Review exercises: Modular powers

1. **Fermat's little theorem.** Compute $5^{100} \bmod 7$.
2. **Euler's theorem.** Compute $2^{100} \bmod 21$.

Can Euler's theorem also be applied to $3^{100} \bmod 21$? Why?

Review exercises: Modular powers

1. **Fermat's little theorem.** Compute $5^{100} \bmod 7$.
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3. **Challenge.** Compute

$$2^{100} \bmod 35.$$

Try both methods:

- Fermat's little theorem and CRT.
- Euler's theorem.

4.

Optional: Miller–Rabin Primality Testing

In memory of Michael Rabin



Gary L. Miller

Knuth Prize, 2013



Michael O. Rabin (1931–2026)

Turing Award, 1976

Photos: Andrej Bauer, 2004 (Miller; Rabin), CC BY-SA 2.5 SI; scaled, otherwise unchanged.

Rabin: IAS memorial; Miller: ACM SIGACT Knuth Prize.

Primality testing

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Given an integer $n > 1$, determine whether n is prime or composite.

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Today: Use fast checks to prove that an input is **composite**.

Fermat's little theorem: The contrapositive

Let $n > 2$ and $1 < a < n$, with $\gcd(a, n) = 1$.

Recall Fermat's little theorem

$$n \text{ prime} \implies a^{n-1} \equiv 1 \pmod{n}.$$

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Warning: The reverse implication is not generally true!

A composite n can still satisfy $a^{n-1} \equiv 1 \pmod{n}$. Passing the check does **not** prove primality.

A Fermat check

Input: An integer $n > 2$
and a base $1 < a < n$.

By FLT (when $\gcd(a, n) = 1$)
$a^{n-1} \not\equiv 1 \pmod{n}$
$\implies$ n is composite

1. $\gcd(a, n) > 1 \implies$ found a **non-trivial divisor** of $n \implies$ n is **composite**.
   Euclid: $O(\log n)$ divisions.

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2. Otherwise compute $r = a^{n-1} \bmod n$ by **repeated squaring**.
   $O(\log n)$ modular multiplications.

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Result	Conclusion
$r \neq 1$	n **is composite.**
$r = 1$	No conclusion.

Examples with base 2

n	$2^{n-1} \bmod n$	Conclusion
15	$4 \neq 1$	**Composite**
$341 = 11 \times 31$	1	No conclusion
17	1	No conclusion

The algorithm only detects **composites**; it never certifies primes.

Strengthening the test

Two ways to detect more composites:

1. Test more bases

Try prime bases $2, 3, 5, \ldots$:

$$2^{n-1}, \quad 3^{n-1}, \quad 5^{n-1}, \ldots \pmod{n}.$$

For $n = 341$, base 3 detects compositeness:

$$3^{341-1} \equiv 56 \not\equiv 1 \pmod{341}.$$

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1. Test more bases

Try prime bases $2, 3, 5, \ldots$:

$$2^{n-1}, \quad 3^{n-1}, \quad 5^{n-1}, \ldots \pmod{n}.$$

2. Use a stronger test

Miller–Rabin Test

Widely used in practice, e.g., OpenSSL.

Helps generate RSA keys for HTTPS—the secure connections your browser uses.

Miller–Rabin: A compositeness test

Let $n > 2$ be odd. A **square root of** 1 **modulo** n satisfies

$$x^2 \equiv 1 \pmod{n}.$$

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Proof. Choose the representative $1 \leq x \leq n-1$. Then

$$n \mid (x^2 - 1) = (x-1)(x+1).$$

If n were prime, it would divide $x-1$ or $x+1$. Since $1 \leq x \leq n-1$, this forces $x = 1$ or $x = n-1$—a trivial root.

This contradicts the choice of x. Thus n is composite.

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Core idea of Miller–Rabin: Look for nontrivial square roots of 1.

$32^2 = 1024 \equiv 1 \pmod{341}$, but $32 \not\equiv \pm 1 \pmod{341}$.

Thus 32 is a nontrivial square root of 1, so 341 **is composite.**

Miller–Rabin: Strengthening Fermat's test

Let $n > 2$ be odd, with $1 < a < n$ and $\gcd(a, n) = 1$.

1. Recall the Fermat check. Compute $a^{n-1} \bmod n$.

$$a^{n-1} \not\equiv 1 \pmod{n} \quad \Longrightarrow \quad n \text{ composite.}$$

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2. If $a^{n-1} \equiv 1 \pmod{n}$, then

$$\left(a^{\frac{n-1}{2}}\right)^2 = a^{n-1} \equiv 1 \pmod{n}.$$

If $a^{\frac{n-1}{2}} \not\equiv \pm 1 \pmod{n}$, we found a nontrivial square root of 1, so n **is composite**.

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Full Miller–Rabin: Search for nontrivial square roots of 1 by testing

$$a^{\frac{n-1}{2}}, \quad a^{\frac{n-1}{4}}, \quad \ldots \pmod{n}.$$

Until the exponent can't be divided by 2 (becomes odd).

Miller–Rabin: Guarantee

Guarantee of the full test

For any odd **composite** $n > 3$, choose a base a uniformly at random from $\{2, 3, \ldots, n-2\}$.

$$\Pr[\text{MillerRabin}(n, a) = \text{“Composite”}] \geq \frac{3}{4}.$$

On the other hand, if n is prime, the test never outputs “Composite”.

Repeat the test t times independently.

The chance of missing a composite drops **exponentially!**

Class 2: Summary

- **Fermat:** $a^{p-1} \equiv 1 \pmod{p}$ when p is prime and $p \nmid a$.
- **Euler:** $a^{\varphi(m)} \equiv 1 \pmod{m}$ when $\gcd(a, m) = 1$.

Optional: Miller–Rabin primality testing

- A failed Fermat check proves compositeness; a passed check does not prove primality.